

# Correction Endpoints — Developer Reference

Base URL: <https://lactosure.azurewebsites.net/api/MachineCorrection>

## 1. Save Correction from App (JSON)

Use this when the Dairy/Society user enters correction values through the web app.

### Endpoint

```
POST /api/MachineCorrection/GetLatestMachineCorrection
Content-Type: application/json
```

### Request Body

```
{
  "societyID": "SOC001",
  "machineType": "LactoScan",
  "machineId": "33",
  "ch1": "1",
  "fat1": "3.50",
  "snf1": "8.20",
  "clr1": "22.00",
  "t1": "0.00",
  "w1": "0.00",
  "p1": "0.00",
  "ch2": "2",
  "fat2": "6.50",
  "snf2": "9.10",
  "clr2": "27.00",
  "t2": "0.00",
  "w2": "0.00",
  "p2": "0.00",
  "ch3": "3",
  "fat3": "5.00",
  "snf3": "8.70",
  "clr3": "24.00",
  "t3": "0.00",
  "w3": "0.00",
  "p3": "0.00"
}
```

| Field     | Required | Description                    |
|-----------|----------|--------------------------------|
| societyID | Yes      | Society / Collection center ID |

| Field                       | Required | Description                                              |
|-----------------------------|----------|----------------------------------------------------------|
| <code>machineType</code>    | Yes      | Model name of the machine                                |
| <code>machineId</code>      | Yes      | Machine ID (numeric or prefixed, e.g. <code>M33</code> ) |
| <code>ch1/ch2/ch3</code>    | No       | Channel labels (default: 1, 2, 3)                        |
| <code>fat1/snf1/clr1</code> | No       | Channel 1 calibration values (Cow)                       |
| <code>fat2/snf2/clr2</code> | No       | Channel 2 calibration values (Buffalo)                   |
| <code>fat3/snf3/clr3</code> | No       | Channel 3 calibration values (Mix)                       |
| <code>t1/t2/t3</code>       | No       | Temperature correction per channel                       |
| <code>w1/w2/w3</code>       | No       | Water correction per channel                             |
| <code>p1/p2/p3</code>       | No       | Protein correction per channel                           |

### What happens internally

- A backup copy is saved to `CorrectionBackups` (full history, never overwritten).
- The live row in `Corrections` is **upserted** — one active row per (`societyID`, `machineId`, `machineType`).

### Success Response

```
{ "message": "Correction saved and backup created successfully." }
```

### Error Responses

| Status | Reason                                                                                   |
|--------|------------------------------------------------------------------------------------------|
| 400    | <code>societyID</code> , <code>machineId</code> , or <code>machineType</code> is missing |
| 500    | Database or server error                                                                 |

## 2. Save Correction from Physical Machine (Pipe-delimited)

Use this when the milk-testing machine pushes its own correction data back to the cloud.

**Endpoint** — supports both GET and POST (form field `InputString`)

```
GET /api/MachineCorrection/SaveMachineCorrectionFromMachine?InputString=<value>
POST /api/MachineCorrection/SaveMachineCorrectionFromMachine (form-data:
InputString=<value>)
```

### InputString Format

```
CollectionCenter|MachineType|Version|MachineId||Channel|Fat|Snf|Clr|Temperature|Water|Protein
```

## Example

```
SOC001|LactoScan|v2|M33||1|F3.50|F8.20|F22.00|F0.00|F0.00|F0.00
```

| Position | Field            | Notes                                            |
|----------|------------------|--------------------------------------------------|
| 0        | CollectionCenter | Society ID                                       |
| 1        | MachineType      | Machine model                                    |
| 2        | Version          | Firmware version                                 |
| 3        | MachineId        | Leading letter stripped automatically (M33 → 33) |
| 4        | (empty)          | Reserved                                         |
| 5        | Channel          | 1/b = Cow · 2/c = Buffalo · 3/d = Mix            |
| 6        | Fat              | Leading letter stripped (F3.50 → 3.50)           |
| 7        | Snf              |                                                  |
| 8        | Clr              |                                                  |
| 9        | Temperature      |                                                  |
| 10       | Water            |                                                  |
| 11       | Protein          |                                                  |

## Channel Routing

| Channel value | Saved to fields                                              |
|---------------|--------------------------------------------------------------|
| 1 or b        | CowFat, CowSnf, CowClr, CowTemperature, CowWater, CowProtein |
| 2 or c        | BufFat, BufSnf, BufClr, BufTemperature, BufWater, BufProtein |
| 3 or d        | MixFat, MixSnf, MixClr, MixTemperature, MixWater, MixProtein |

## Validation

- Society must exist in [CollectionCenterAdds](#).
- Machine must exist in [MachineDetailss](#) with matching [MachineID](#) and [ModelName](#).

## Success Response

```
"Machine correction saved successfully."
```

## Error Responses

| Status | Reason                                                                    |
|--------|---------------------------------------------------------------------------|
| 400    | Society not found — "Failed to save correction. Invalid token."           |
| 400    | Machine not found — "Failed to save correction. Invalid machine details." |
| 400    | Parse or format error                                                     |

## 3. Get Correction History (All Records)

Use this to list every correction ever saved for a specific society + machine combination. Returns records from [CorrectionBackups](#), ordered newest first.

### Endpoint

```
GET /api/MachineCorrection/GetCorrectionByDetails?SocietyID=<value>&MachineID=
<value>&MachineType=<value>
```

### Query Parameters

| Parameter                   | Required | Description                    |
|-----------------------------|----------|--------------------------------|
| <a href="#">SocietyID</a>   | Yes      | Society / Collection center ID |
| <a href="#">MachineID</a>   | Yes      | Machine ID                     |
| <a href="#">MachineType</a> | Yes      | Machine model name             |

### Example Request

```
GET
https://lactosure.azurewebsites.net/api/MachineCorrection/GetCorrectionByDetails?
SocietyID=SOC001&MachineID=33&MachineType=LactoScan
```

### Success Response — 200 OK

```
[
  {
    "id": 12,
    "societyID": "SOC001",
    "machineId": "33",
    "machineType": "LactoScan",
```

```
"backupDateTime": "2026-05-06T10:30:00",
"ch1": "1", "fat1": "3.50", "snf1": "8.20", "clr1": "22.00", "t1": "0.00",
"w1": "0.00", "p1": "0.00",
"ch2": "2", "fat2": "6.50", "snf2": "9.10", "clr2": "27.00", "t2": "0.00",
"w2": "0.00", "p2": "0.00",
"ch3": "3", "fat3": "5.00", "snf3": "8.70", "clr3": "24.00", "t3": "0.00",
"w3": "0.00", "p3": "0.00",
"status": 1
}
]
```

Error Responses

| Status | Reason                                                                        |
|--------|-------------------------------------------------------------------------------|
| 400    | Any of <b>SocietyID</b> , <b>MachineID</b> , or <b>MachineType</b> is missing |
| 404    | No correction history found for the given combination                         |
| 500    | Server error                                                                  |

**Note:** Every time a correction is saved via Endpoint 1, a snapshot is appended here. This list is never overwritten — it is the full audit trail.

Quick Comparison

|                    | Endpoint 1 (App — Save)                                         | Endpoint 2 (Machine — Save)            | Endpoint 3 (History — Fetch)    |
|--------------------|-----------------------------------------------------------------|----------------------------------------|---------------------------------|
| Caller             | Flutter web app / developer                                     | Physical milk-testing machine          | Flutter web app / developer     |
| Method             | POST (JSON)                                                     | GET or POST (pipe string)              | GET (query params)              |
| Table written/read | <b>Corrections</b> (upsert) + <b>CorrectionBackups</b> (append) | <b>SaveMachineCorrections</b> (append) | <b>CorrectionBackups</b> (read) |
| History kept       | Yes — every save creates a backup row                           | Yes — every push is a new row          | —                               |
| Channels           | All 3 in one request                                            | One channel per request                | All channels returned           |